ETH zürich

Literature Review

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March 11, 2026

Overview of literature review

- Literature review of 13 papers
- Main topics: crack propagation in snow, snow modeling, modeling in more general sense (introduction to different modeling methods used)
- Modeling methods: Material Point Method (MPM) and Discrete Element Method (DEM)

Modeling Methods

- Two main methods used: MPM and DEM
- DEM described by Cundall and Strack, 1979, material of disks & spheres, contact forces and resulting displacements calculated
- MPM where material is discretized in points with Eulerian background mesh to calculate stresses at points and time-dependent equations, for example in Gaume et al., 2018
- MPM is more favorable for slope-scale simulations because of modelling of grain contacts
- Other models for snow such as SNOWPACK (Bartelt & Lehning, 2002) and CROCUS (Brun et al., 1992) model snowpack response to external factors

Avalanche Release - General Crack Propagation

- All papers agree on avalanche formation by loading - weak layer failure
- Some papers discuss anticrack propagation (closure of cracks formed) like eg Gaume et al., 2018
- Others state shear cracks to be the main failure mode
- Bergfeld et al., 2025 combine the two main hypotheses: anticrack early in avalanche propagation, shearing later

Bibliography

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